

Subarray Sum Constraints

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to construct an array $x_1, x_2, \dots, x_n$ consisting of n integers.

The array must satisfy m constraints of the form (l, r, s) : the sum $x_l + x_{l+1} + \dots + x_r$ must equal s .

Input

The first line has two integers n and m : the length of the array and the number of constraints.

The next m lines each have three integers l, r and s : the description of the constraints.

Output

If a solution exists, print YES on the first line.

On the second line, print n integers $x_1, x_2, \dots, x_n$: the contents of the array. All elements of the array must satisfy $-10^{15} \leq x_i \leq 10^{15}$ and the array must satisfy all given constraints. You can print any valid solution.

If no solution exists, just print NO.

Constraints

- $1 \leq n \leq 5000$
- $0 \leq m \leq 2 \cdot 10^5$
- $1 \leq l \leq r \leq n$
- $-10^9 \leq s \leq 10^9$

Example

Input	Output
5 3 1 3 3 3 5 3 4 4 -1	YES 0 2 1 -1 3